

Hyojung Han

AI Systems Engineer

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SUMMARY

- AI systems engineer with 19 years of continuous experience (2007–present) spanning computer vision, recommendation systems, generative AI, and enterprise AI platforms.
- Currently leads product strategy and architecture for three AI platforms at ThakiCloud: inference, training, and enterprise agent automation.
- Personally trains and measures Korean 27B alignment, on-device compression, and serving quantization, shipping 34 open models on HuggingFace and 5 sole-authored arXiv papers.
- Founded and ran a generative AI content platform that reached 50K paying users and KRW 400M in cumulative revenue across Japan, Taiwan, and Hong Kong, and runs a bilingual technical blog with 1,328 published posts.

CORE TECHNICAL SKILLS

Agent Harness & Orchestration Harness Engineering · Loop Engineering · Graph Engineering · MCP Tool Integration · Skill Routing / RAG · LangGraph · Human-in-the-Loop Approval · Policy & Audit Trail · Deterministic Verification Gates · Multi-Agent Consensus

Training, Fine-tuning & Distillation SFT · CPT · DPO · GRPO · GKD · LoRA / QLoRA · Knowledge Distillation · FSDP / DDP · TRL · Unsloth · MLflow · Eval Sets & Regression Testing

Inference, Serving & Compression vLLM · NVIDIA Dynamo · LMCache · Prefill-Decode Disaggregation · NVFP4 / W4A16 · LLM Compressor · MoE Compression / Pruning · Scale-to-Zero · KServe · Model Routing

Speech, Vision & Generative Models VoxCPM2 TTS · Qwen3-TTS · Qwen3-ASR · Speaker Diarization (pyannote) · VLM Video Understanding · Stable Diffusion · DreamBooth · GPT Image · Text-to-Video · EmbeddingGemma · AdaFace / ArcFace · OpenCV / ViT

Platform & GPU Infrastructure Kubernetes · Kueue / MultiKueue · ArgoCD / GitOps · Helm · Keycloak · SeaweedFS · CloudNativePG · NATS · Prometheus / Grafana · CUDA · Docker

Languages & Frameworks Python · Go (Fiber) · TypeScript · C++ · React 19 · Next.js · FastAPI · PyTorch · PostgreSQL · Redis

Classical ML & Optimization LightGBM · CatBoost · XGBoost · scikit-learn · Google OR-Tools · Airflow · Linear Programming / Constraint Optimization · Time Series / Ranking

PROFESSIONAL EXPERIENCE

ThakiCloud AI Platform Strategy, Architecture & Team Lead Seoul, South Korea Apr 2025 ~ Present

Leads strategy, architecture, and delivery across three AI platforms, and personally trains and measures the Korean alignment, on-device compression, and serving-quantization work behind 34 open model releases and 5 papers.

Led strategy and architecture for three AI platforms

- Set product direction and core architecture for inference, training, and agent automation, and led the engineering team by setting experiment priorities
- Directed GPU scheduling, multi-tenancy, and GitOps deployment on Kubernetes, and personally wired the agent builder into internal systems

Korean 27B response-style alignment and its off-target effects (Human-KO)

- LoRA-SFT aligned Qwen3.8-27B for Korean response style: bullet usage 97.3% to 0.6%, median length 1,329 to 152 characters, human-likeness win rate 1.1% to 94.9% (n=175); the model only moved once the training set was cut from 2,992 records to 519 curated ones
- Found that style alignment shifts behaviors the objective never targeted, then corrected them with DPO: KoBBQ ambiguous-context abstention 81.5% to 92.8%, stereotyped answers 16.1% to 6.7%, with KMMLU up 4.2pp

Korean dialect model and a measurement trap in synthetic data (Satoori-KO)

- Built KoDialectBench (1,000 items, 5 regions, 3 axes) and trained the 27B on both real and synthetic lanes
- Quantified how a metric inflates results when its scoring inventory overlaps the synthetic-construction inventory: on a construction-disjoint arm, measured dialect recovery collapses from 91.8% to 8.1% while pipeline-independent metrics barely move

Compressing the agent skill-retrieval embedder (SKILLRET-Edge)

- Distilled a 0.6B teacher into 22M and 109M students, raising NDCG@10 from 44.37 to 75.26 and putting the skill-router embedder on CPU with no GPU
- Quantized to 17.07MB (int4/g16) while holding NDCG at 75.14, statistically tied with fp32; the 109M student retains 98.0% of teacher quality

2i Studio (Founder) **Founder** Yongin, South Korea Mar 2024 ~ Apr 2025

Founded and ran a generative AI content platform end to end, from product to engineering to operations.

Built a face-based generative AI content platform

- Designed an identity-preserving generation pipeline on DreamBooth fine-tuning, growing to KRW 400M cumulative revenue and 50K paying users

Toss **Senior Machine Learning Engineer** Seoul, South Korea Jan 2021 ~ Feb 2024

Designed and shipped machine learning systems across CDP personalization, generative AI tooling, face authentication, and workforce scheduling.

Customer data platform and personalization prediction

- Built per-service, per-user CTR/CVR prediction models, replacing manual SQL segment queries with automated probability scores

Samsung Electronics **Senior Software Engineer, ML Researcher, Development Manager** Suwon, South Korea

Sep 2011 ~ Dec 2020

Built an ML platform, 5G network automation, and a music recognition engine, while also shaping the organization's technical direction.

Built an internal cloud machine learning platform

- Designed a Kubernetes-based ML platform, cutting model deployment lead time from days to hours

Daum **Machine Learning Engineer** Jeju, South Korea Sep 2008 ~ Aug 2011

Started in computer vision, building object recognition and image search systems.

Object recognition and image search system

- Designed a real-time image matching system at 7M-image scale using histogram-based Hamming embedding and LIS indexing

Vision Startup **Software Engineer** Seoul, South Korea Jan 2007 ~ Aug 2008

Began my career in image processing and pattern recognition, the groundwork for later image search systems.

Image processing and pattern recognition modules

- Implemented C++ vision modules that extracted and matched features from camera input

PUBLICATIONS

Sole-authored papers from work trained and measured in-house, each with reported numbers and reproduction code. Alongside them, 34 trained and compressed models are released under the HuggingFace ThakiCloud organization across five collections: Human-KO, Satoori-KO, SKILLRET-Edge, vocabulary pruning, and serving quantization.

Off-Target Effects of Response-Style Alignment in a Korean 27B Language Model

arXiv:2609.11291 (Sep 2026)

On-Device Commercial Intent Retrieval Under Size, Latency, and Privacy Constraints: A 3 MiB Retrieval System with Typed Egress Boundaries

arXiv (submitted)

Rewrite Locality and Cache Provenance: Two Axes That Govern Prefix Reuse in Multi-Turn LLM Agents

arXiv (submitted)

Metric-Construction Coupling Inflates Measured Synthetic Dialect Recovery

arXiv (submitted)

Where Post-Training Quantization Breaks Text Embedders: A Measured Map Across Four Embedder Families

arXiv (submitted)

EDUCATION

Yonsei University M.S. in Computer Science 2007